

# Chyi Ricketts

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## EDUCATION

### Imperial College London

London, United Kingdom

September 2026

*Master of Research in Cancer Informatics*

- **Key Project:** Novel Biomarker Discovery for Mesenchymal Glioblastomas via Multi-Omics Analysis
- **Thesis:** Target-Specific Binding Affinity Prediction Scoring Functions for Drug Target Identification in Lead Optimization

### Duke University & Duke Kunshan University (DKU) Dual Degree

May 2025

Durham, North Carolina, USA & Kunshan, Jiangsu, China

*Bachelor of Science in Molecular Bioscience with a Track in Genetics and Genomics*

- **Honors:** *Graduation with Distinction* | GPA: 3.7/4.0
- **Thesis:** Integrating Segmentation Software and BLIP-2 for Automated Cellular Image Analysis in a Web-Based Application

## LABORATORY SKILLS

### Wet Lab Techniques

Mammalian and bacterial cell culture  
DNA/RNA extraction and purification  
Prokaryotic siRNA production and purification  
PCR and RT-qPCR  
Western blotting  
CRISPR/Cas9 gene editing  
Confocal & fluorescence microscopy  
Zebrafish embryo microinjection

### Computational and Bioinformatics Tools

**Python** (data analysis, bioinformatics, ML)  
**R / RStudio** (statistical analysis, visualization)  
**Other Languages:** SQL, HTML/CSS, Java, JS  
**Workflow & reproducibility:** Git/GitHub, Docker, Nextflow  
**Sequence analysis:** MMseqs2, TM-align, BLAST, bwa, kallisto  
**Machine learning:** scikit-learn, PyTorch, TensorFlow  
**Image analysis:** ImageJ/Fiji, Cellpose, Segment Anything  
**Systems:** Linux (command line), HPC (PBS/qsub), AWS

## RELEVANT RESEARCH EXPERIENCE

### Imperial College London ~ *Ballester Lab*

Nov. 2025 – Sep. 2026

London, United Kingdom

#### Project: Multi-Omics Identification of a Novel Prognostic Biomarker in Glioblastoma Subtypes

- Applied high-throughput multi-omics approaches for biomarker discovery in TCGA glioblastoma cohorts, integrating gene expression, methylation, CNV, proteomic, and clinical data to identify prognostic molecular signatures

#### Conducting Masters thesis research on ML scoring functions for cancer drug discovery across five protein targets

- Curated and preprocessed large-scale protein–ligand datasets (~100,000 complexes) with RDKit and ECFP6 Tanimoto FPs
- Designed a benchmarking framework evaluating ~100 protein-ligand models across featurization schemes (AEV-PLIG, ECIF, PLEC) and learning algorithms (GATv2, XGBoost, Random Forest, Support Vector Machine)
- Characterized a generalization-specificity trade-off in BAP scoring functions for descriptor-based models and GNNs

### Duke Kunshan University ~ *Huang Lab*

May 2023 – May 2025

Kunshan, Jiangsu, China

#### Project: RNAi screen to identify novel genes involved in cancer cell proliferation

- Streamlined the extraction and purification of prokaryotic siRNA followed by transfection of siRNA into HeLa cells to evaluate the extent of protein expression and toxicity

**Grant writing:** Authored and submitted a successful grant proposal to the Kunshan Supercomputing Center, resulting in the award of 9 high-performance computing nodes to support image analysis and segmentation software benchmarking

#### Thesis: Computational benchmarking of image segmentation tools for biological microscopy

- Conducted quantitative benchmarking of 8 image analysis tools, including ML/DL segmentation methods
- Developed a web-based tool for interactive visualization and analysis of post-segmentation image outputs
- Trained models and evaluated segmentation performance using custom computational metrics
- Performed supporting RNAi-based gene knockdown experiments in HeLa cells for functional validation

### Duke University ~ *Di Talia Lab & Bagnat Lab*

April 2022 – June 2022

Durham, NC, USA

#### Project: Investigation of protein participation in the interstitial lumen of zebrafish (<https://doi.org/10.1242/dev.199994>)

- Applied Knock-in Tagging methods to examine protein participation in packaging and transporting nutrients in the interstitial lumen of zebrafish through embryo injection, electrophoresis gels, PCR, and fluorescent microscopy screening
- Handling of live zebrafish for tail clips for mutant line maintenance
- Maintained laboratory stocks and upheld standard microbiology aseptic techniques

#### Project: DST (Duke Science and Technology) SPARK Project: “Synchronized Clocks in Zebrafish Patterning”

- Trained 10 Cellpose models and quantitatively evaluated their performance using custom Python metrics

## ADDITIONAL SKILLS

- **Languages:** Native English Speaker, Fluent in Mandarin and Advanced literacy in Traditional and Simplified Chinese